

02 Sky-Harvest

Overview

A fruit and vegetable rooftop cultivation and drone delivery system that aims to shorten the distance from "farm to table".

Theme

Urban Space Reuse
Agricultural Planting
Drone Delivery

01 Background

Urban residents face delays and inefficiencies in the fresh food supply chain, with long transport times leading to a decline in the freshness and flavor of fresh produce. Currently, there are several major pain points for urban consumption:

Low transport efficiency & prolonged delivery cycles

lead to reduced freshness, compromised flavor, and diminished nutrient content by the time produce reaches consumers.

High vulnerability Urban Supply Chain

such as extreme weather events, transport strikes, or policy changes, which can quickly impact availability and pricing.

Dietary inequalities

in urban areas, where lower-income neighborhoods are dominated by unhealthy options like fast food and ultra-processed snacks, crowding out access to fresh produce.

Over-reliance on centralized retail channels

creates bottlenecks and limits flexibility in supply, making the systems slow to respond to changing demand.

No personalized planting or product selection mechanisms

leaving consumers unable to influence crop types, harvest timing, or product variety according to their preferences and needs.

02 Questionnaire

Omaid Mustafa

Personal Info

Male
28 Years Old
Local New Yorker / UX designer
Knows almost everything about NYC
Enjoy Watching Movies / Eat Brunch

Pain Points

Buying groceries in the city is not easy. I often have to commute to go to Brooklyn to get to one of the stores which is most cost-worthy. I think how to keep and know the freshness of things I am purchasing is essential.

Common Services



Gabbi Xue

Personal Info

Female
29 Years Old
RTO employee in NYC
Living in NYC for few months already
Enjoy Eating Salad, Vegetables

Pain Points

As a temporary resident, there are too many choices for supermarkets in the city, but I often find it hard to buy and cook fresh various veggie or fruits to make my own personal diet meal.

Common Services



Lara Bennett

Personal Info

Female
43 Years Old
Landscape Designer
Live in NJ, work on the island
Have a unique taste in Bonsai

Pain Points

I love to travel a lot and enjoy trying out new things. For instance, I really like the Yubari Melon while I was traveling in Hokkaido, Japan. But it's impossible or too rare to find them in US. It would be awesome to have them here.

Common Services



Jing Bin

Personal Info

Male
52 Years Old
Senior designer in China
Traveling in NYC for 2 weeks
Like to try different fruit tarts

Pain Points

I think the best way to know a place is trying out their local foods, but if I want to cook or get something I eat back home, it is always hard to find fresh materials.

Common Services



03 Research

Manhattan As Test Ground

Manhattan's extreme density ($\approx 70,000$ people/sq mi) creates concentrated demand for fresh food, making it a prime location to pilot scalable solutions like rooftop farms or drone delivery that could maximize limited space and serve large populations efficiently.

Age Distribution



Ethnic Comparisons



Manhattan is the ideal test ground for VertiFresh due to its high population density, diverse cultural preferences, fast-paced lifestyle.

Summary

Manhattan faces many challenges in accessing fresh from the earth products due to



long-distance supply chains



limited availability of user customized options



inconvenience of traditional retail

Related Projects

amazon fresh

Model: Unmanned fresh food vending station
Reasons for failure
Rent costs account for 45% of revenue
Freshness complaint rate of 32%
Theft and damage



Model: Community bulk ordering + designated delivery point

Reasons for failure
Minimum order quantity requirements led to the loss of users
Last-mile delivery costs even more

Model: Large commercial building with two floors Rooftop farm

Reasons for failure
The price was still 25% higher
failed network effects



04 Solution Development

Problem Statement

Manhattan residents face challenges due to longdistance supply chains, limited access to fresh and diverse producetime constraints for in-person shopping, and rapid loss of freshness during transportation, impacting community health,dietary quality, and environmental sustainability.

DEFINE

- Last-mile delivery for perishables is inefficient.
- Demand for fresher, tastier options is rising.
- Urban diets lean unhealthy due to poor access
- Overreliance on manuallabor slows delivery.
- Freshness declines rapidly in current systems

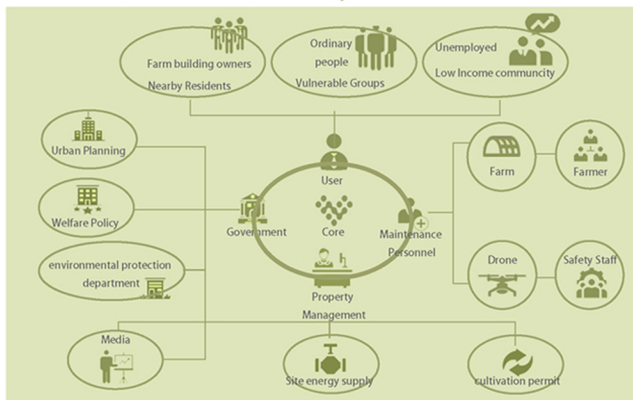
IDEATE

- Rooftop farms ensure freshness
- Drone delivery with smart routing
- Vertical charging towers for quick deployment
- Airspace "areen corridors" for food routes
- Cuts transport steps, lowering carbon emissions.

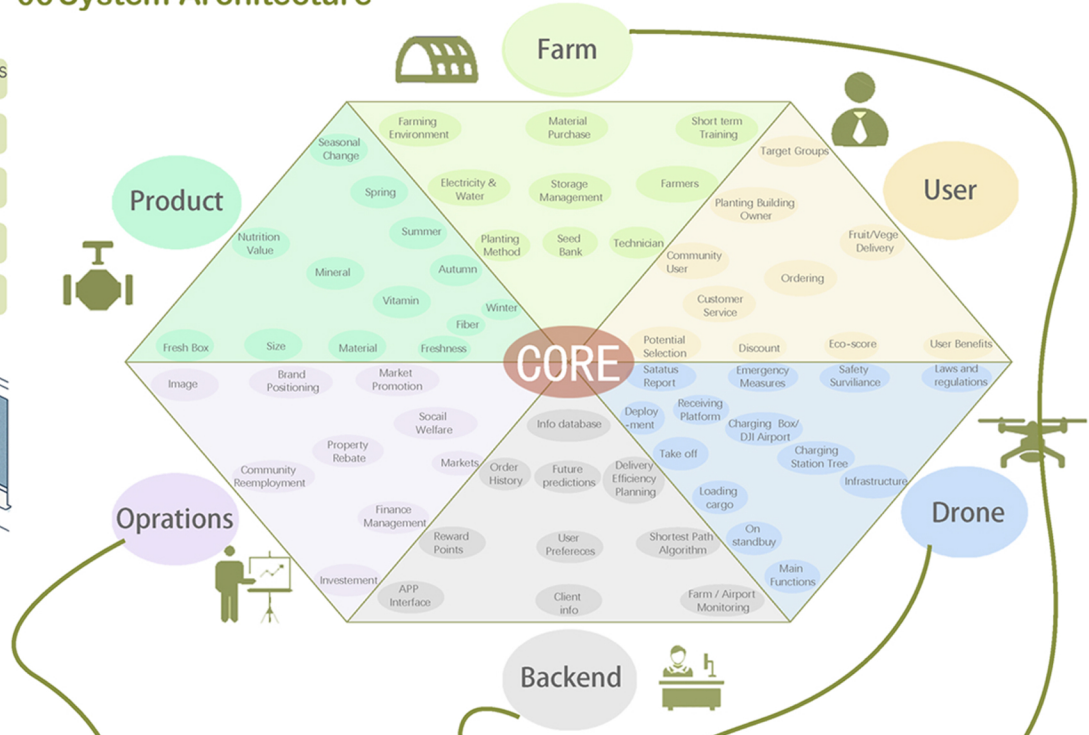


For high-rise window height restrictions, the delivery can be resolved by installing additional self-service parcel lockers on the ground

05 Stakeholders Map



06 System Architecture



07 Bussiness Model Canvas

Key Partnerships

- Building/estate owners with rooftop access
- Drone operators & logistics agencies
- Local government for airspace & safety regulations
- Urban agriculture suppliers

Key Activities

- Rooftop farm setup & maintenance
- Drone delivery & route optimization
- Customer order management
- Crop planning & monitoring

Key Resources

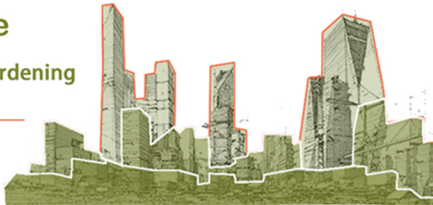
- Rooftop spaces & equipment
- Drone fleet & charging stations
- Farm management software & AI tools
- Skilled technicians & operators

Cost Structure

- Farm installation & upkeep
- Drone purchase & maintenance
- Technology infrastructure
- Labor & training

08 Farm & Agriculture

Feasibility Analysis of Roof Gardening

	
	High-rise buildings
	Middle-level buildings
	Low-rise buildings
	Low-rise buildings (1-5 stories)
	Middle-level buildings (6-20 floors)
	High-rise buildings (20 floors or more)
sunlight	Relatively insufficient sunlight
wind	low wind speed
temperature	Stable temperature
	Adequate sunlight
	moderate wind
	Small temperature fluctuations
	Adequate sunlight
	Strong winds
	large temperature differences

Facility Composition

Greenhouse

Greenhouse structures are primarily composed of a steel pipe frame, steel supports, covering materials, and polycarbonate panels.



Drone

Multi-Layer Drone Deployment Tower

A drone automatic deployment and charging tower composed of multiple DJI Dock modules stacked vertically.

Management and Dispatch Facilities

The dome-shaped building serves as the central management and dispatch facility for the entire system.



09 Facility Coverage Planner

Data Source

Geoson.io Satellite Map
Extracted high resolution satellite images to capture the whole potential farm deployment area, and export it into Geoson file



Potential Available Sites



Employment Linkage

The staff of farms advocate linking unemployed residents to subsidized rooftop gardening and drone safety monitoring roles, where community service hours earn points redeemable for fresh-produce-creating a sustainable cycle of workforce, and mutual aid.

Implementation Support

Eco-Points: For Staff

Earn points through planting and related activities, redeemable for discounts, while tracking cost and eco-impact compared to traditional methods.



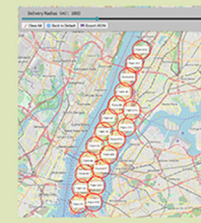
Improvement Process

First few tryouts in coordinating the map with the expected location the radius kept messing up



Final Outcome

It takes 19 farms to cover the whole area of Manhattan considering drone delivery radius as 1km



Eco-Points: For Users

Earn points by returning boxes or recycling promoting community participation and showing clear environmental and cost benefits.



10 Fruit box

The fruit box is made from environmentally friendly, biodegradable materials.



11 Drone Delivery System

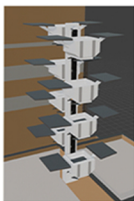
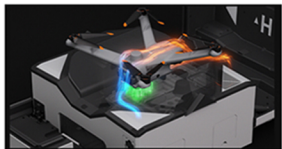
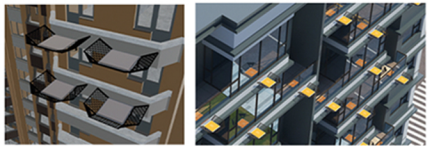
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DJI Matrice 3D
Autonomous GPS-guided drone with obstacle avoidance and a custom mount for secure fresh box delivery.
- 

DJI Airport 3
Fast-charge docking station (20min) with weather monitoring and data sync.
- 

Rotating Charging Station
Wireless, dual-mode charging base for multiple drones, 40% more efficient.
- 

Receiving Platform
Simple steel structure at windows/balconies for verified, secure drop-offs.



- Safety Measures**
- 

Comply with Manhattan drone laws-height limits, no-fly zones, and privacy rules.
- 

Use DJI O3 live feed for real-time monitoring.
- 

Sensor dashboard tracks battery, motor, and GPS status.
- 

A dedicated safety monitor oversees system status at all times.
- 

Apply geofencing to block restricted areas.
- 

Enable one-click emergency takeover for instant control.



12 Methodology

A* Shortest Delivery Route

Data Source

Input

A predefined 10x10 grid map that simulates a drone navigation area. 0 → Free space (drone can fly over it) 1 → Obstacle (e.g., rooftop/building, drone must avoid) in python

```
grid = [
    [0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0],
    [0, 1, 1, 0, 1, 0, 0, 0, 1, 0, 0],
    [0, 1, 0, 0, 1, 0, 0, 0, 1, 0, 0],
    [0, 1, 0, 1, 1, 0, 0, 0, 0, 0, 0],
    [0, 1, 0, 0, 0, 0, 0, 1, 1, 1, 0],
    [0, 1, 0, 1, 1, 1, 0, 0, 0, 1, 0],
    [0, 0, 0, 1, 0, 0, 1, 0, 0, 0, 0],
    [1, 1, 0, 0, 0, 1, 1, 0, 1, 0],
    [0, 0, 0, 1, 0, 0, 0, 0, 0, 1, 0],
    [0, 1, 0, 1, 0, 1, 1, 0, 0, 0, 0]
]
```

Elements Setup

heapq → Manages the open set (priority queue) efficiently.

math → Provides Euclidean distance as the heuristic.

matplotlib, numpy → Used for visualizing the final path.

```
python
import heapq
import math
import matplotlib.pyplot as plt
import numpy as np
```

Pathfinding Core

The core is to find the shortest route from a start point to a goal on a grid.

It combines the actual distance traveled (g) with an estimated distance to the goal (h) using Euclidean distance. It checks eight directions (including diagonals) and avoids visited or blocked cells. The path with the lowest total cost (f = g + h) is expanded first, continuing until the goal is reached or no path is found.

```
def heuristic(x1, y1, x2, y2):
    return math.sqrt((x2-x1)**2 + (y2-y1)**2)

def is_valid(x, y):
    return 0 <= x < 10 and 0 <= y < 10 and grid[x][y] == 0

def is_in_heap(x, y):
    return (x, y) in open_set

def is_in_closed(x, y):
    return (x, y) in closed_set

def is_in_path(x, y):
    return (x, y) in path

def a_star(grid, start, goal):
    open_set = []
    closed_set = []
    path = []

    heapq.heappush(open_set, (0, start))

    while open_set:
        f, (x, y) = heapq.heappop(open_set)

        if (x, y) == goal:
            path.append((x, y))
            return path

        if is_in_closed(x, y) or is_in_path(x, y):
            continue

        closed_set.add((x, y))
        path.append((x, y))

        for dx, dy in [(1, 0), (-1, 0), (0, 1), (0, -1), (1, 1), (1, -1), (-1, 1), (-1, -1)]:
            nx, ny = x + dx, y + dy
            if is_valid(nx, ny) and not is_in_closed(nx, ny) and not is_in_path(nx, ny):
                g = f + 1
                h = heuristic(nx, ny, goal)
                f = g + h
                heapq.heappush(open_set, (f, (nx, ny)))

    return None
```

First Visualization

- **Starting point**: Grayish white
- **End point**: Dark gray
- **Path**: Medium gray
- **Buildings**: Black (impassable)

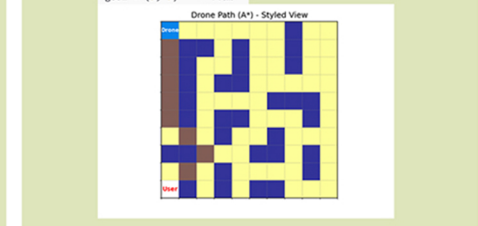


Final Visualization

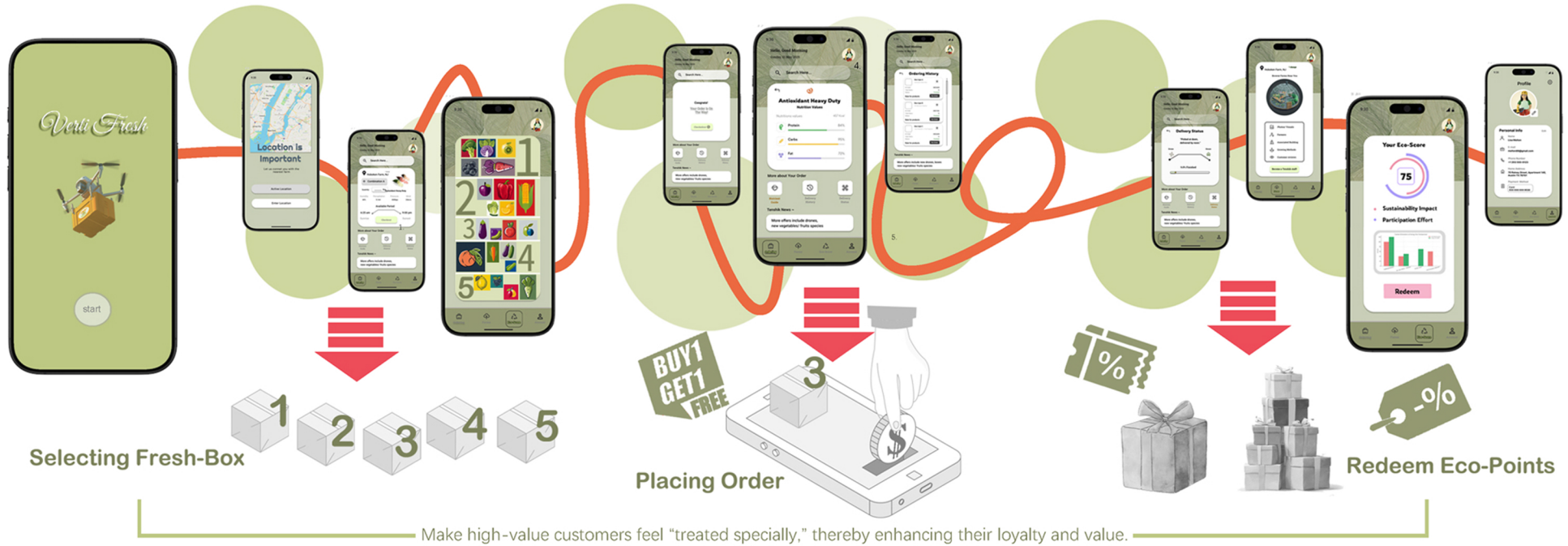
s) and avoids visited or blocked cells. The path with the lowest total cost (f = g + h) is expanded first, continuing until the goal is reached or no path is found.

```
start = (0, 0) # 起点 goal = (9, 9) # 终点
goal = (9, 9) # 终点
```

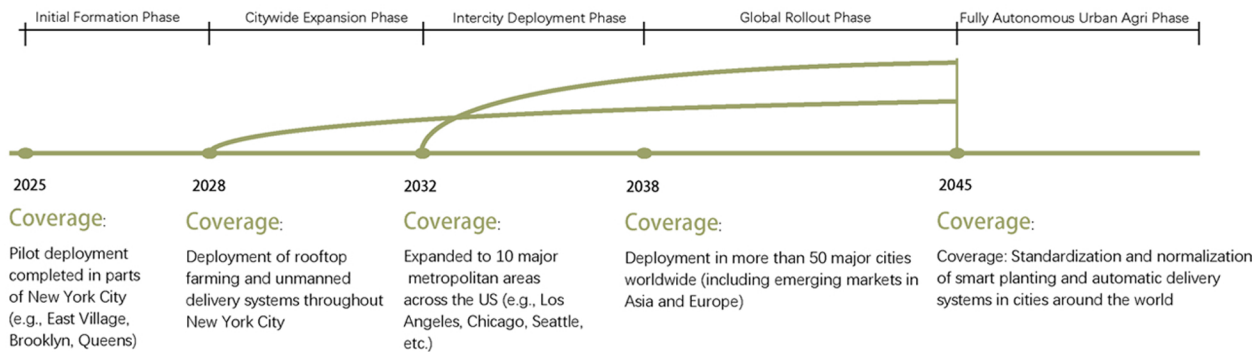
Drone Path (A*) - Styled View



13 APP Interface



14 Future Time Lapse



15 Sustainability and Future Improvement

Infrastructure & Technology ~ Charging Station Upgrades ~ Photovoltaic power supply integration ~ Technology Iteration and Range Optimization ~ Extreme Weather Adaptability	Operations & Coordination ~ Drone Delivery and Charging System ~ Policy and Airspace Coordination
Environmental & Resource Management ~ Rooftop agriculture and energy recycling ~ Improve resource recycling ~ Soil and planting system optimization	